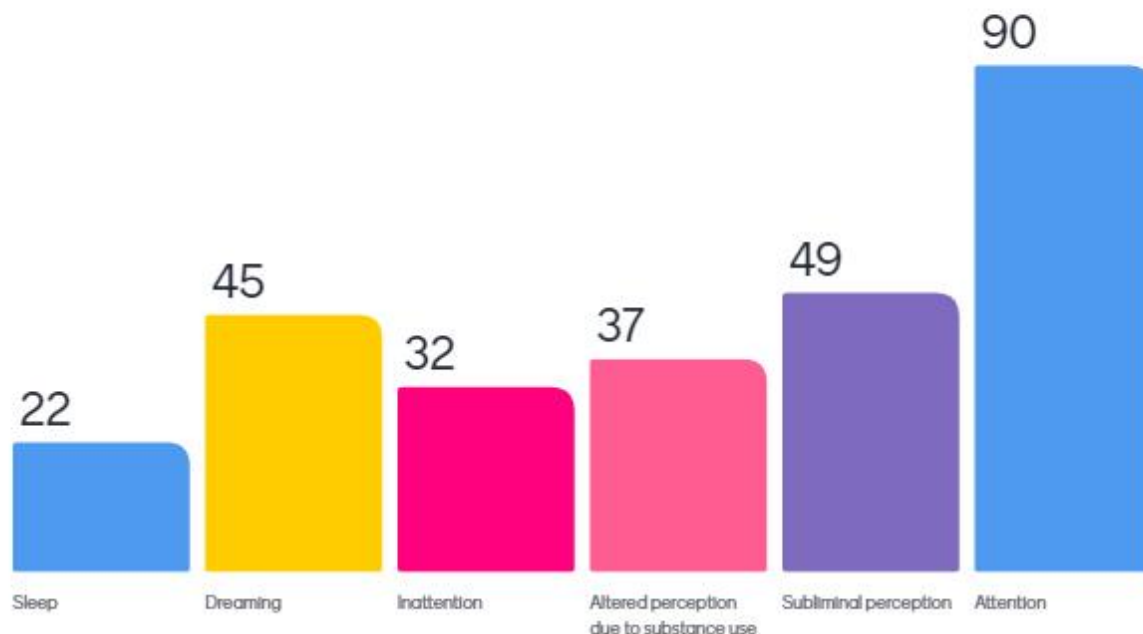

Topic 3

Consciousness

The **hard problem of consciousness**

“the challenge of explaining how and why subjective experiences, or *qualia*, arise from physical brain processes.”

Sensory experiences, feelings, and thoughts have an inner, personal quality that is difficult to fully explain through objective scientific measures alone.



Overview

- Introduction
- Variations in consciousness
- Attention
- Sleep
- Drugs

History of Consciousness

- Psychology began as a science of consciousness.
- Behaviorists alienated consciousness from psychology.
- However, after 1960, mental concepts (consciousness) started reentering psychology.
 - Thanks to information processing technologies!

Consciousness

- Subjective awareness of ourselves and our environment
- An awareness of the sensations, thoughts, and feelings that one is attending to at a given moment.
- More controlled/focused when learning a complex concept or behavior
- Becomes automatic by practice

Subjective awareness of ourselves and our environment

Some states occur spontaneously

Daydreaming

Dreaming

Some are physiologically induced

Hallucinations

Food or oxygen starvation

Some are psychologically induced

Sensory deprivation

Meditation

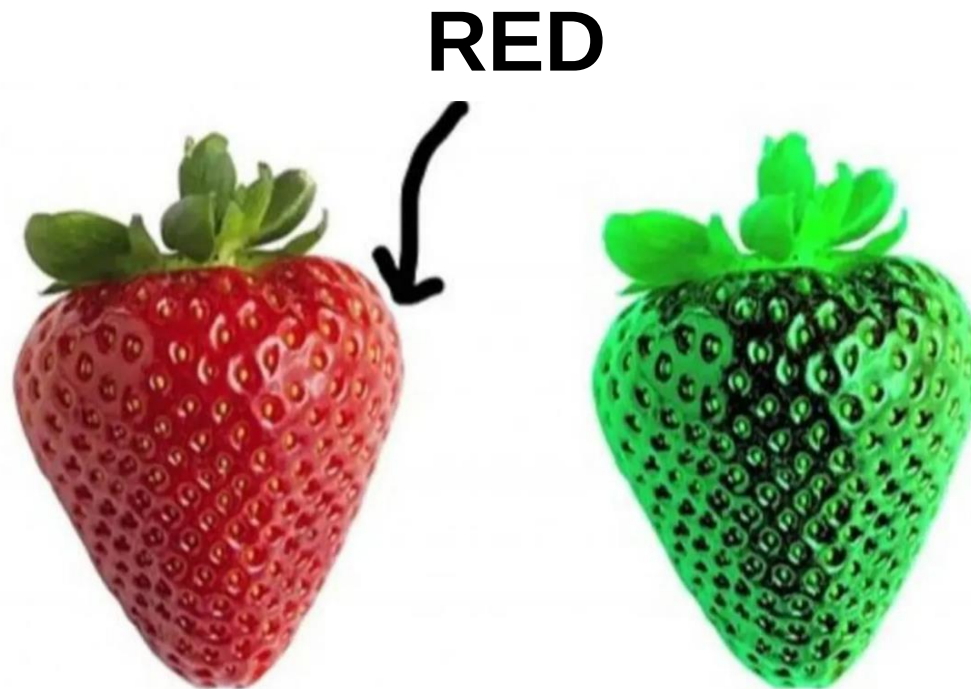


Myers/DeWall, *Psychology*, 13e, © 2021 Worth Publishers
INSADCO Photography/Alamy

Introduction

- **Consciousness:** one's subjective experience of the world, resulting from brain activity
 - The brain and the mind are inseparable.
 - Each of us experiences consciousness personally.
 - We cannot know if two people experience the world in exactly the same way: **explanation gap**

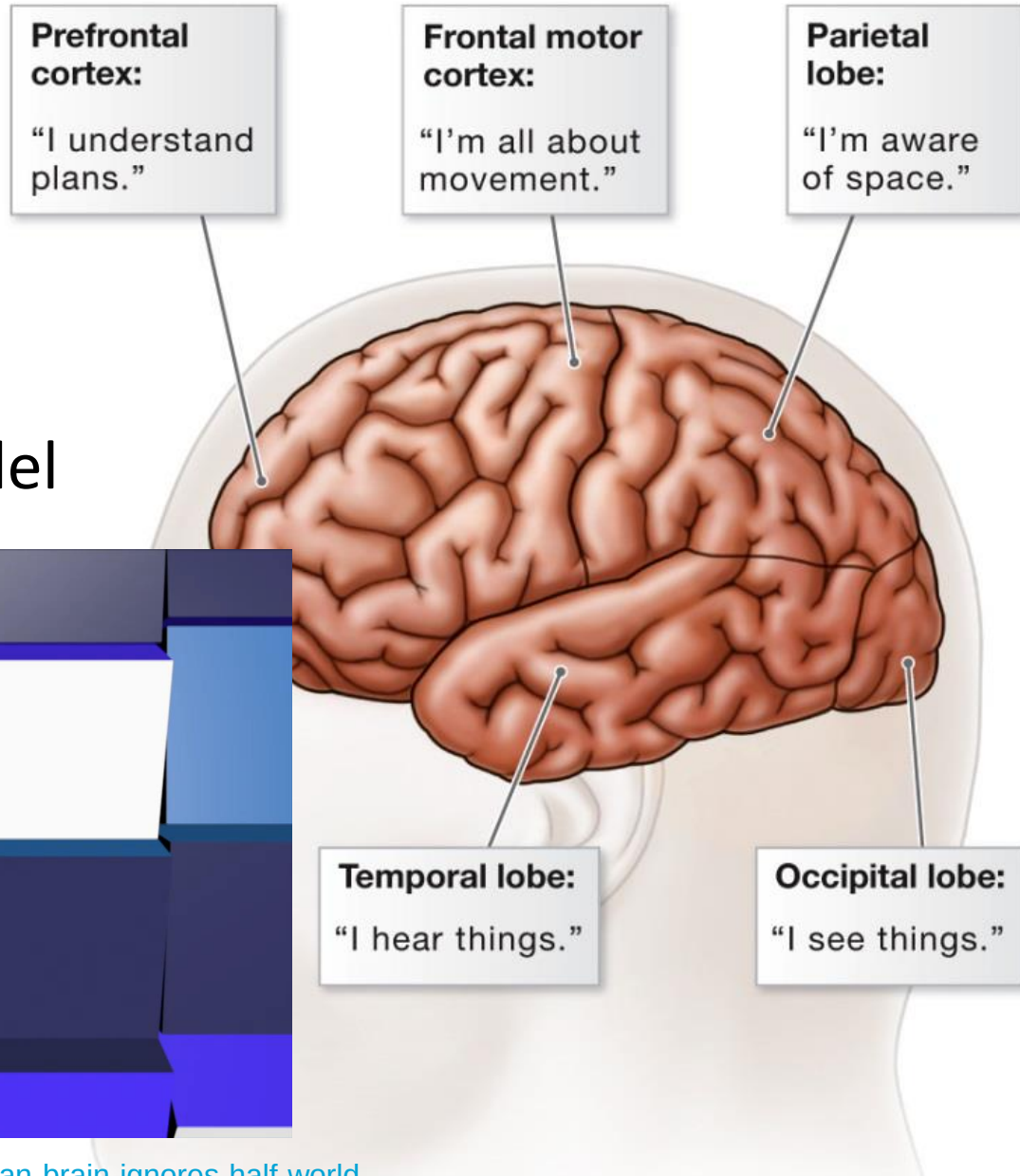
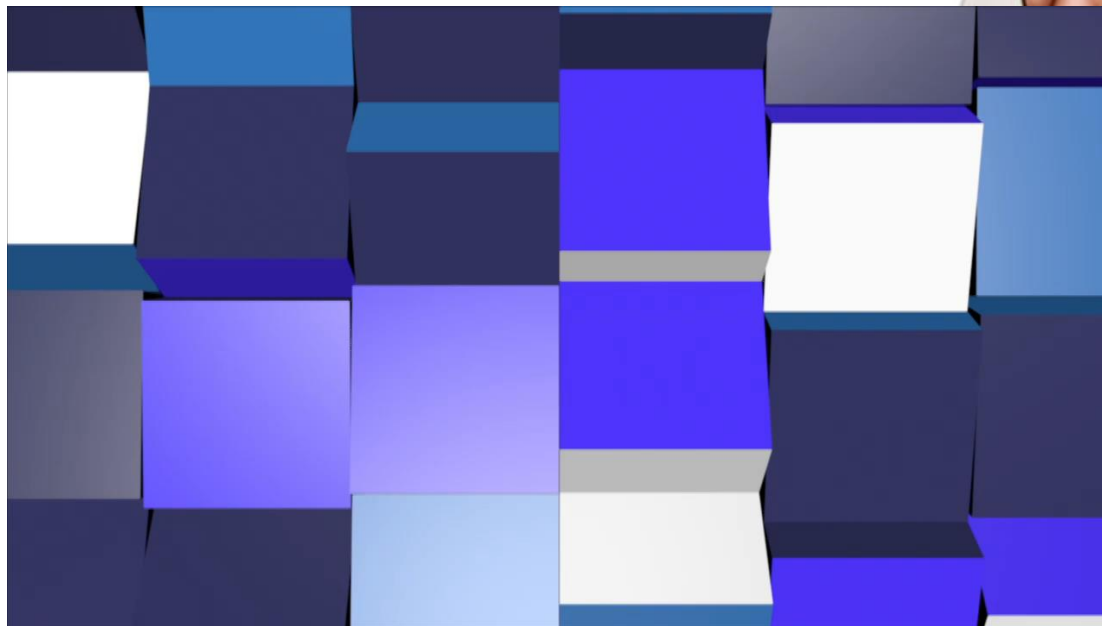
QUALIA



Qualia are the subjective or qualitative properties of experiences. What it feels like, experientially, to see a red rose is different from what it feels like to see a yellow rose.

Introduction

Consciousness arises as a function of which brain circuits are active
-> Global Workspace Model

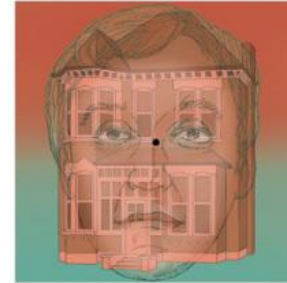


Introduction

HYPOTHESIS: Specific patterns of brain activity can predict what a person is seeing.

RESEARCH METHOD:

- 1 Participants were shown images with houses superimposed on faces.
- 2 Participants were asked to report whether they saw a house or a face.
- 3 Researchers used fMRI to measure neural responses in participants' brains.



RESULTS: Activity increased in the fusiform face area when participants reported seeing a face, but activity increased in temporal cortex regions associated with object recognition when participants reported seeing a house.

CONCLUSION: Type of awareness is related to which brain region processes the particular sensory information.

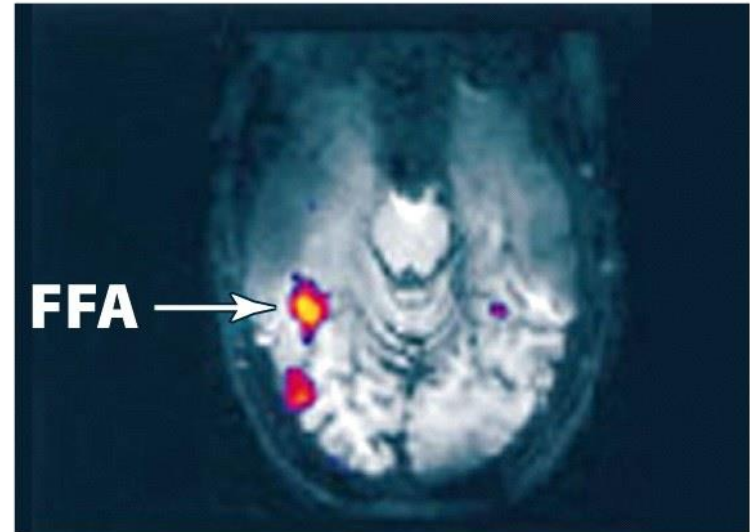
SOURCE: Tong, F., Nakayama, K., Vaughan, J. T., & Kanwisher, N. (1998). Binocular rivalry and visual awareness in human extrastriate cortex. *Neuron*, 21, 753–759.

Introduction

(A)



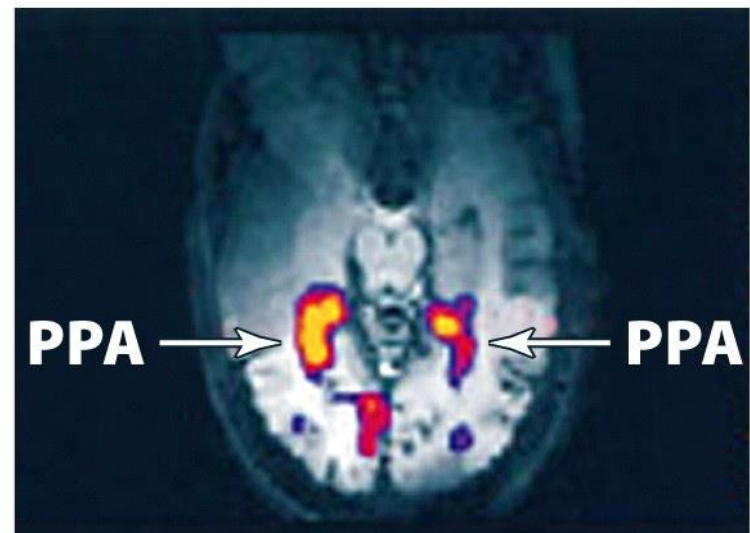
Face



(B)

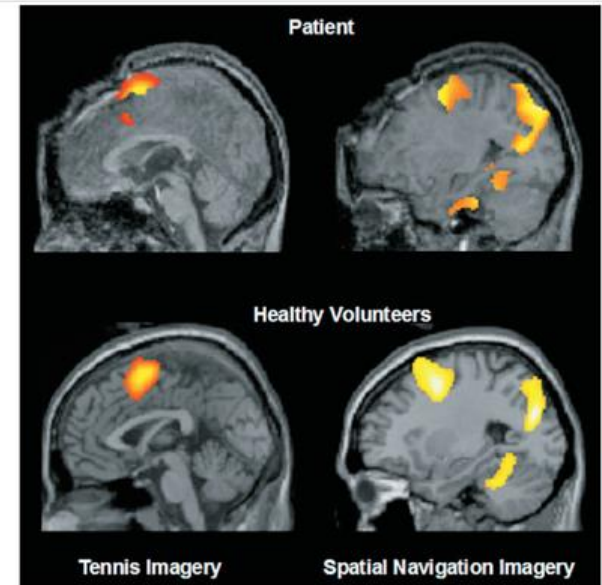


House



Cognitive Neuroscience

- If you think about throwing a basketball, the areas of the brain involved in the planning are activated.
 - Responsiveness in comatose patients

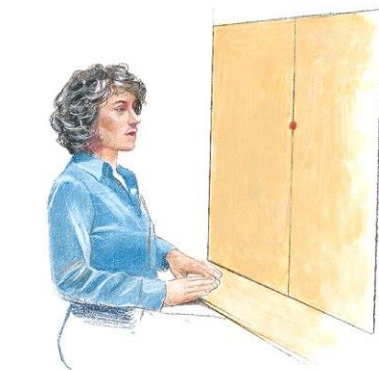
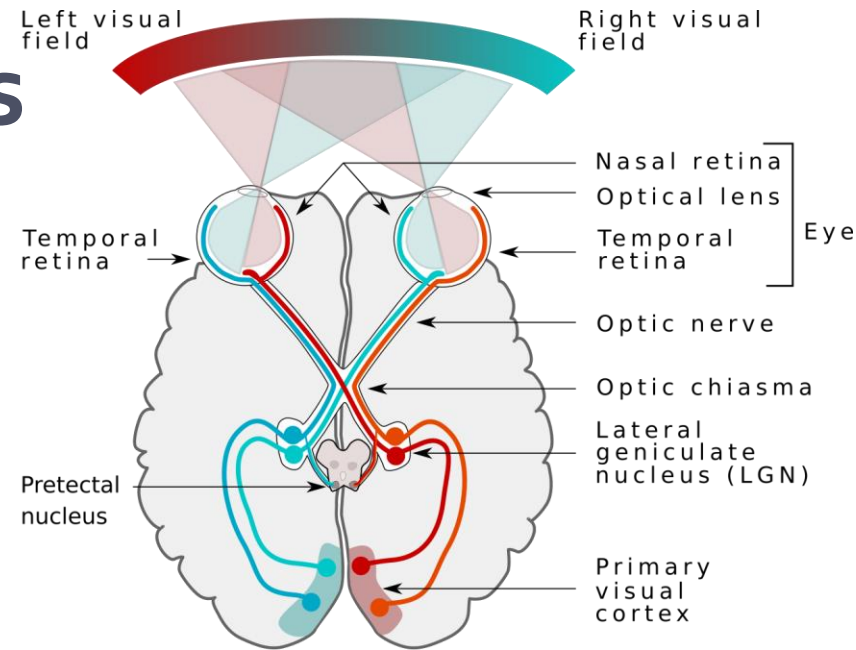
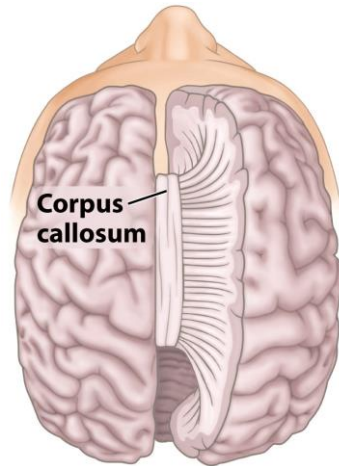


Courtesy of Adrian M. Owen, the Brain and Mind Institute, Western University

Split-Brain Experiment (Sperry & Gazzaniga)

- Subjects were presented information to one or the other side of their brains.
- **Left** hemisphere can tell what it has seen, right hemisphere can show it.
- Studies of split-brain patients:
 - a. Present a picture to the right visual field (left brain)
 - Left hemisphere can tell you what it was.
 - Right hand can show you, left hand can't.
 - b. Present a picture to the left visual field (right brain)
 - Subject will report that they do not know what it was.
 - Left hand can show you what it was, right can't.

Divided Consciousness



“Look at the dot.”



Two words separated by a dot are momentarily projected.

“What word did you see?”

or

“Point with your left hand to the word you saw.”



FIGURE 2.15 Myers/DeWall, *Psychology*, 12e, © 2018 Worth Publishers Andrew Swift

The brain

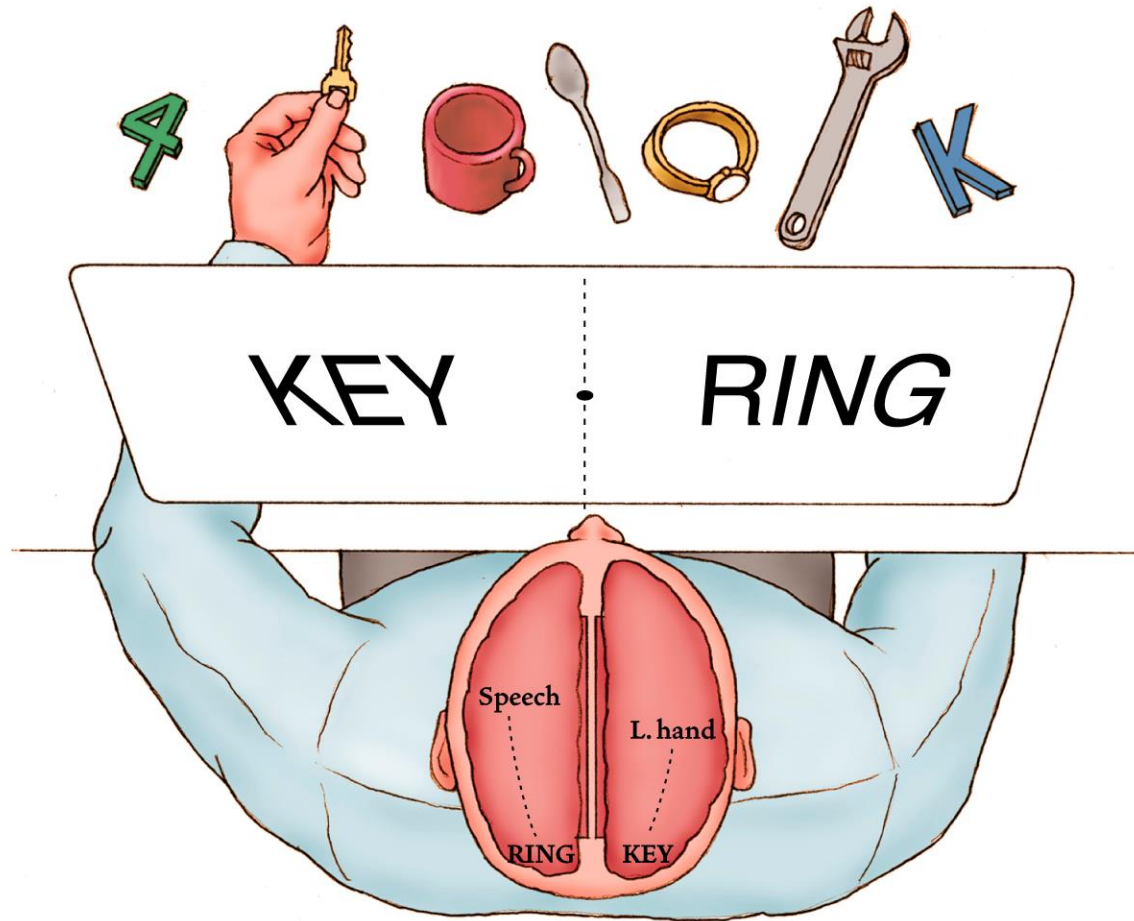
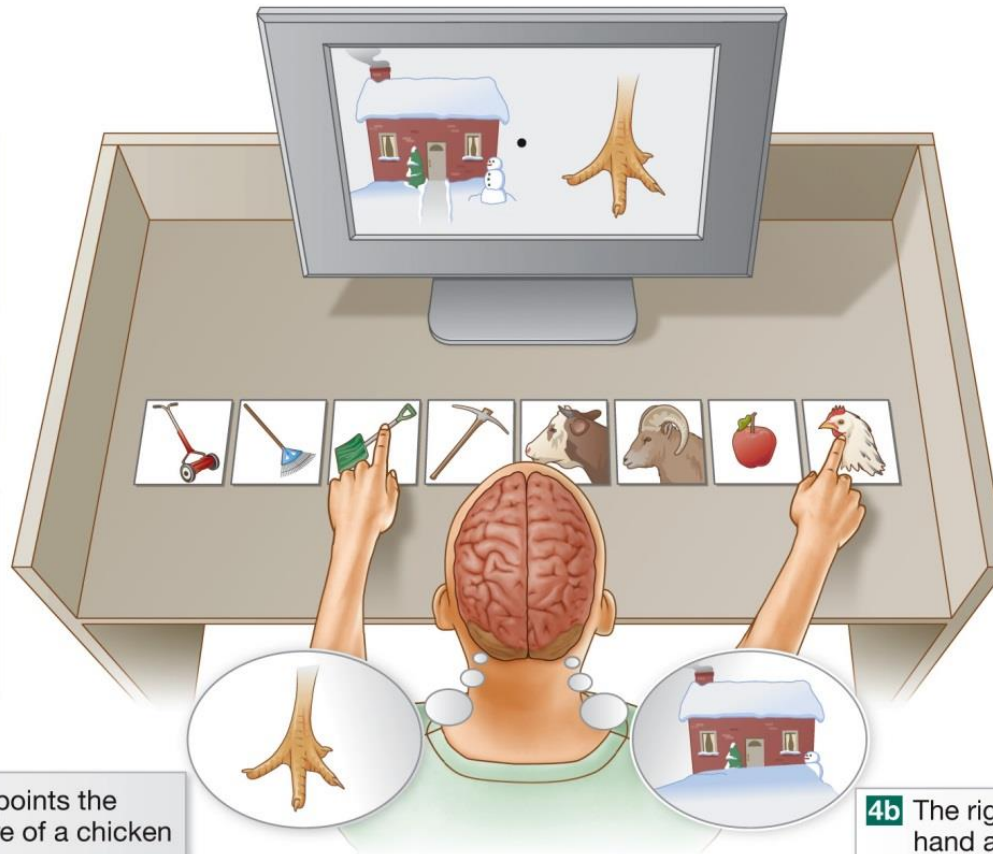


Figure 2.18: A setup sometimes used in split-brain studies

1 A split-brain participant watches as different images flash simultaneously on the left and right.

2 Below the screen is a row of other images.

3 The patient is asked to point each hand at a bottom image most related to the image flashed on that side of the screen.



4a The left hemisphere points the right hand at a picture of a chicken head.

4b The right hemisphere points the left hand at a picture of a snow shovel.

5 When the split-brain participant is asked to explain these selections, the verbal left hemisphere provides the answers. To explain the right hand's selection of the chicken head, the left hemisphere says that the chicken claw goes with the chicken head. To explain the left hand's selection of the shovel, the left hemisphere must interpret, because it does not see the snow scene. The left hemisphere decides that the shovel is used to clean up after chickens.

Psychological Review

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VOLUME 84 NUMBER 3 MAY 1977

Telling More Than We Can Know: Verbal Reports on Mental Processes

Richard E. Nisbett and Timothy DeCamp Wilson
University of Michigan

Evidence is reviewed which suggests that there may be little or no direct introspective access to higher order cognitive processes. Subjects are sometimes (a) unaware of the existence of a stimulus that importantly influenced a response, (b) unaware of the existence of the response, and (c) unaware that the stimulus has affected the response. It is proposed that when people attempt to report on their cognitive processes, that is, on the processes mediating the effects of a stimulus on a response, they do not do so on the basis of any true introspection. Instead, their reports are based on a priori, implicit causal theories, or judgments about the extent to which a particular stimulus is a plausible cause of a given response. This suggests that though people may not be able to observe directly their cognitive processes, they will sometimes be able to report accurately about them. Accurate reports will occur when influential stimuli are salient and are plausible causes of the responses they produce, and will not occur when stimuli are not salient or are not plausible causes.

Examiner: (*indicating patient's left hand*) What is this over here? Take a look over here. What is this?

Patient: Your fingers

Examiner: Take a good look. What is it? (*taps the back of patient's hand*)

Patient: The back of your hand.

Examiner: The back of my hand?

Patient: Yes.

Examiner: Suppose I told you that this was your hand?

Patient: I wouldn't believe you.

Examiner: You wouldn't believe me?

Patient: No, no.

Examiner: This is your hand.

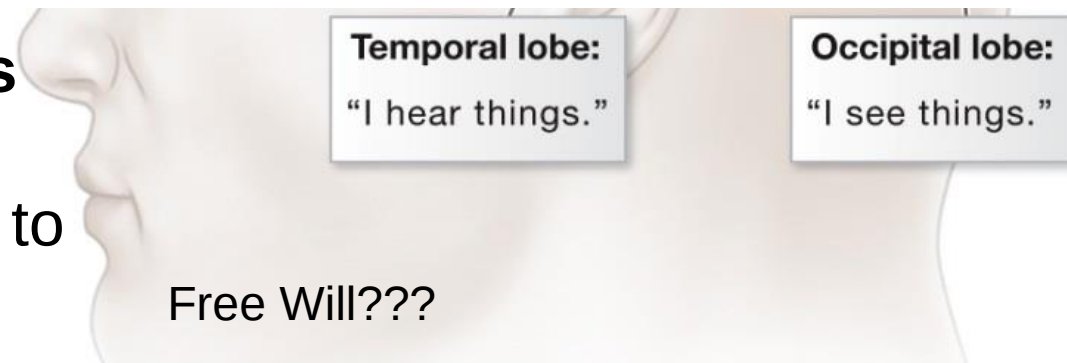
Patient: No.

<https://www.tandfonline.com/doi/full/10.1080/13546805.2011.564925>

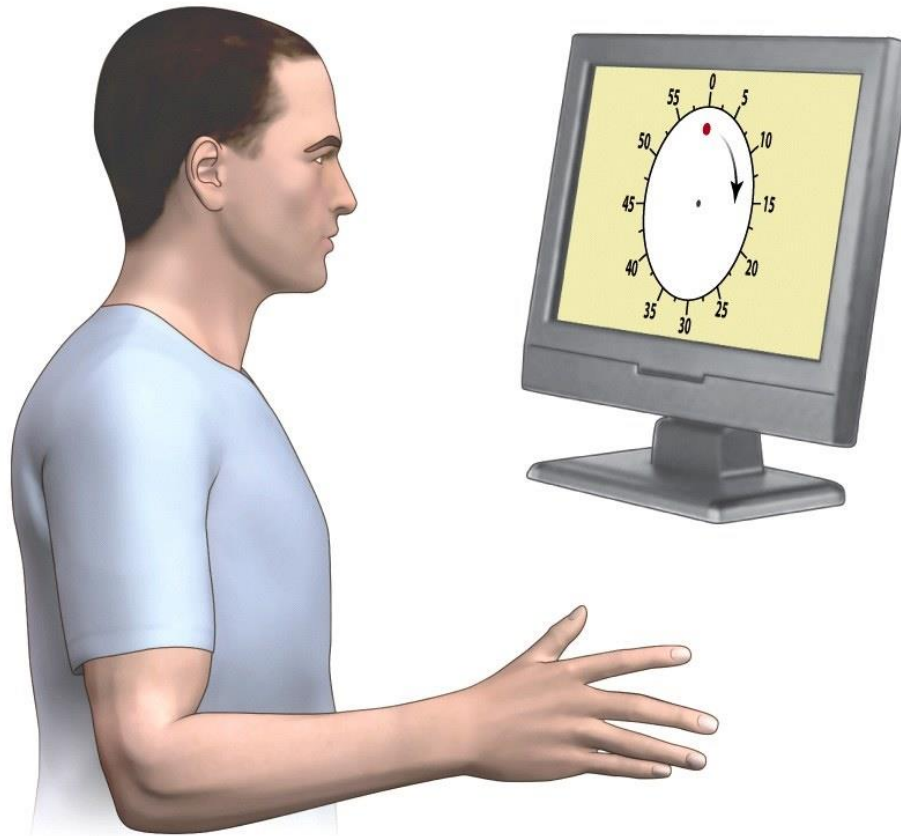
Somatoparaphrenia: Feinberg, Deluca, Giacino, Roane, and Solms (2005, pp. 103–104)

After-the-fact-explanations

- unconscious acts that the conscious mind struggles to explain



Introduction



Variations in consciousness

- Variations in “normal” consciousness
 - Attention
 - Sleep
 - Drugs

Attention



THE ATTENTION TEST

Attention

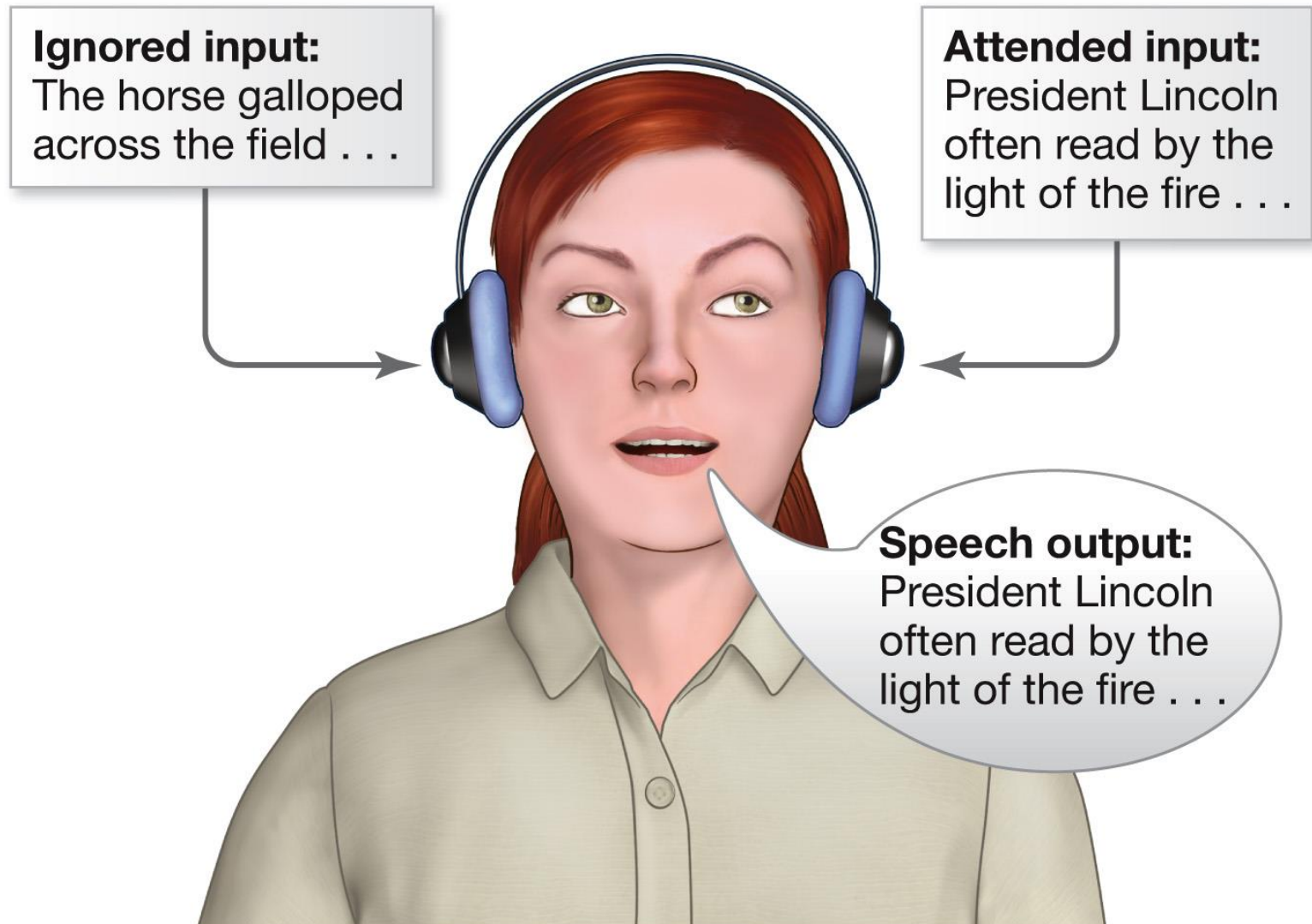
- Attention refers to the process that enables you to focus selectively on some things and avoid focusing on others



Reprinted with permission of Bill Whitehead

- Rapid toggling between activities is the enemy of sustained, focused attention!

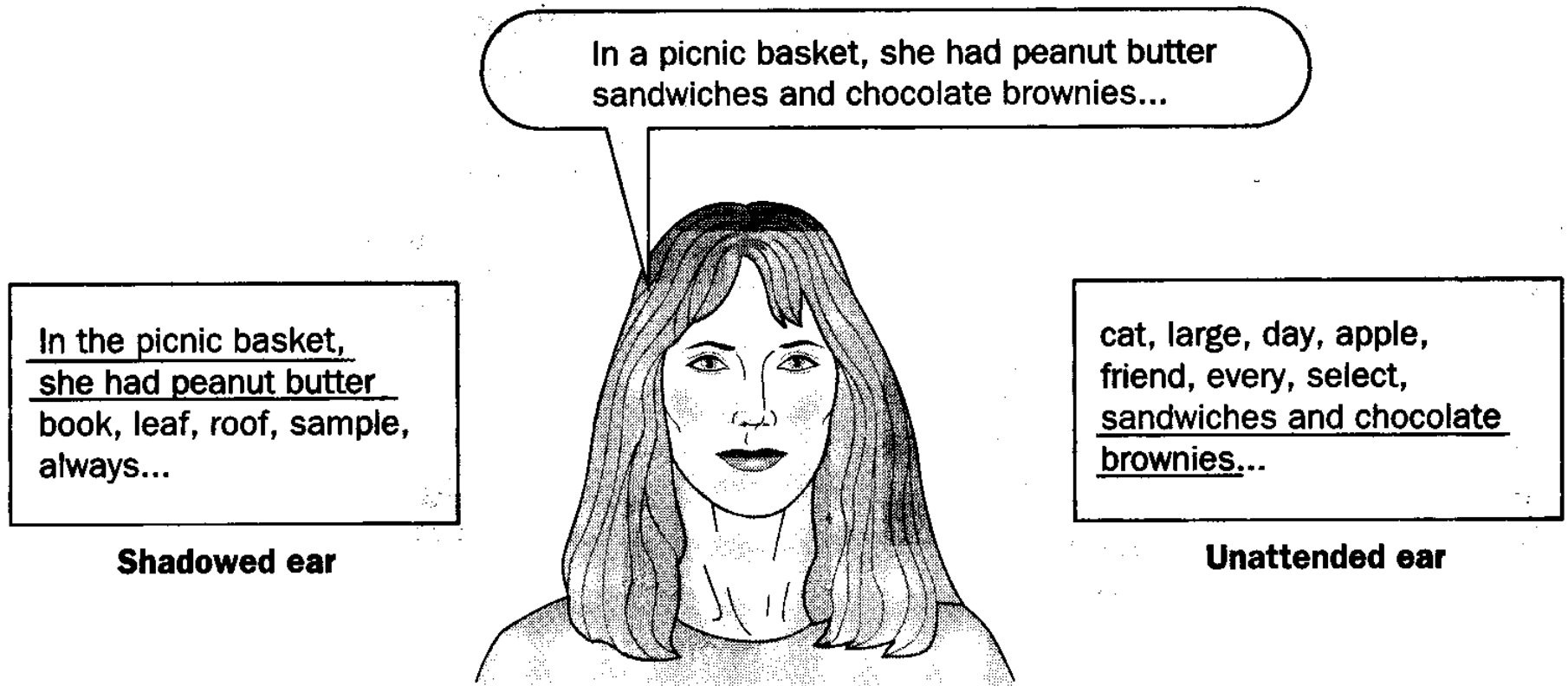
Attention - Dichotic listening (Shadowing)



Attention

- Attention refers to the process that enables you to focus selectively on some things and avoid focusing on others
- But:
 - Cocktail party phenomenon (Cherry, 1953)
 - Sometimes the unattended information breaks through
 - The ability to attend selectively to only one voice among many.

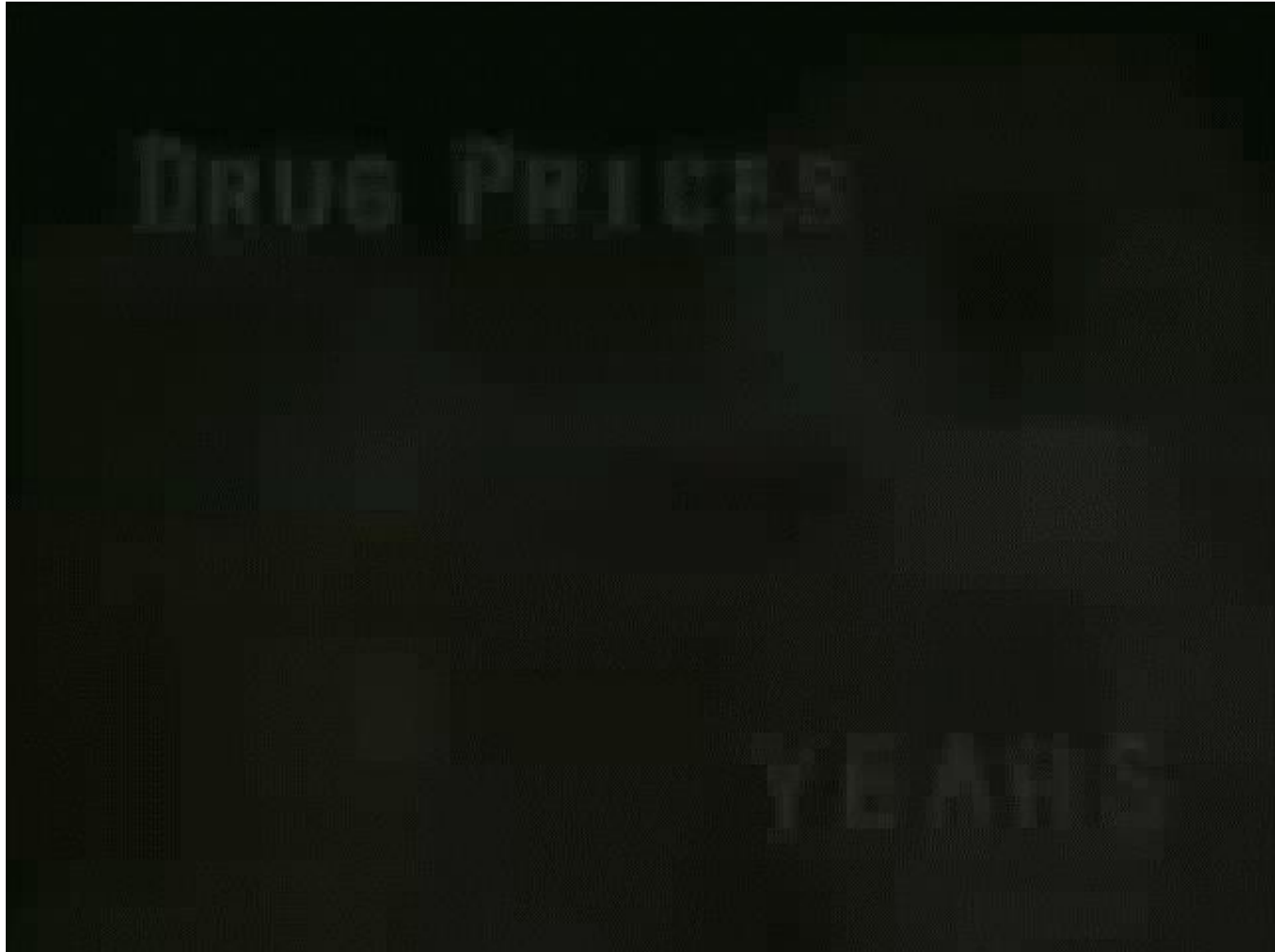
Attention – Dichotic Listening



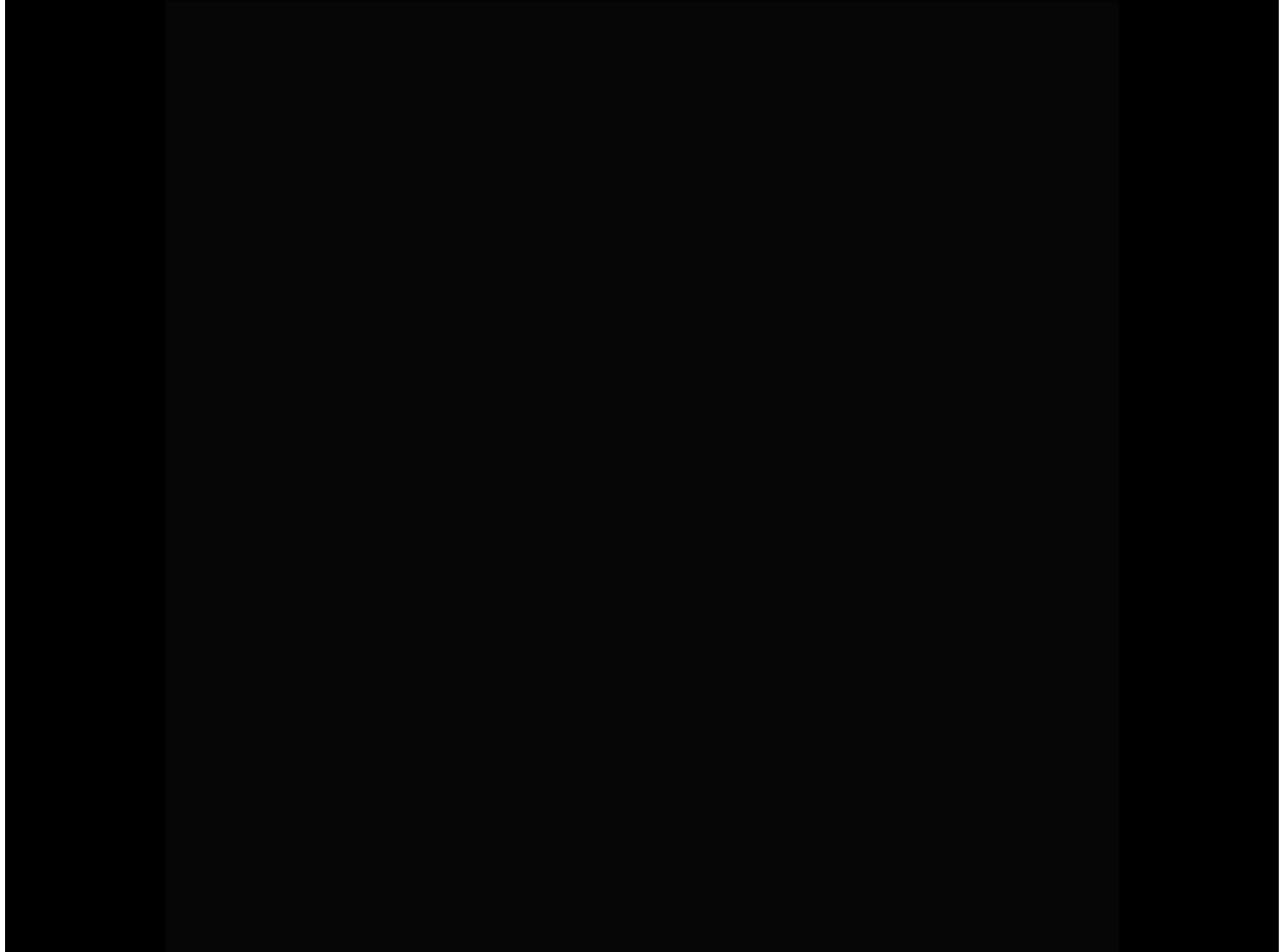
Attention

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 - Unattended information may sometimes affect behavior -> subliminal perception (minimally!)

Attention



Attention



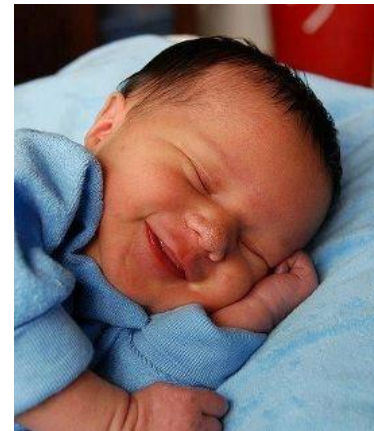
Attention

- Attention refers to the process that enables you to focus selectively on some things and avoid focusing on others
 - Cocktail party phenomenon (Cherry, 1953)
 - Sometimes the unattended information breaks through
 - The ability to attend selectively to only one voice among many.
 - Unattended information may sometimes affect behavior -> subliminal perception (minimally!)
 - Yet, without much awareness -> change blindness

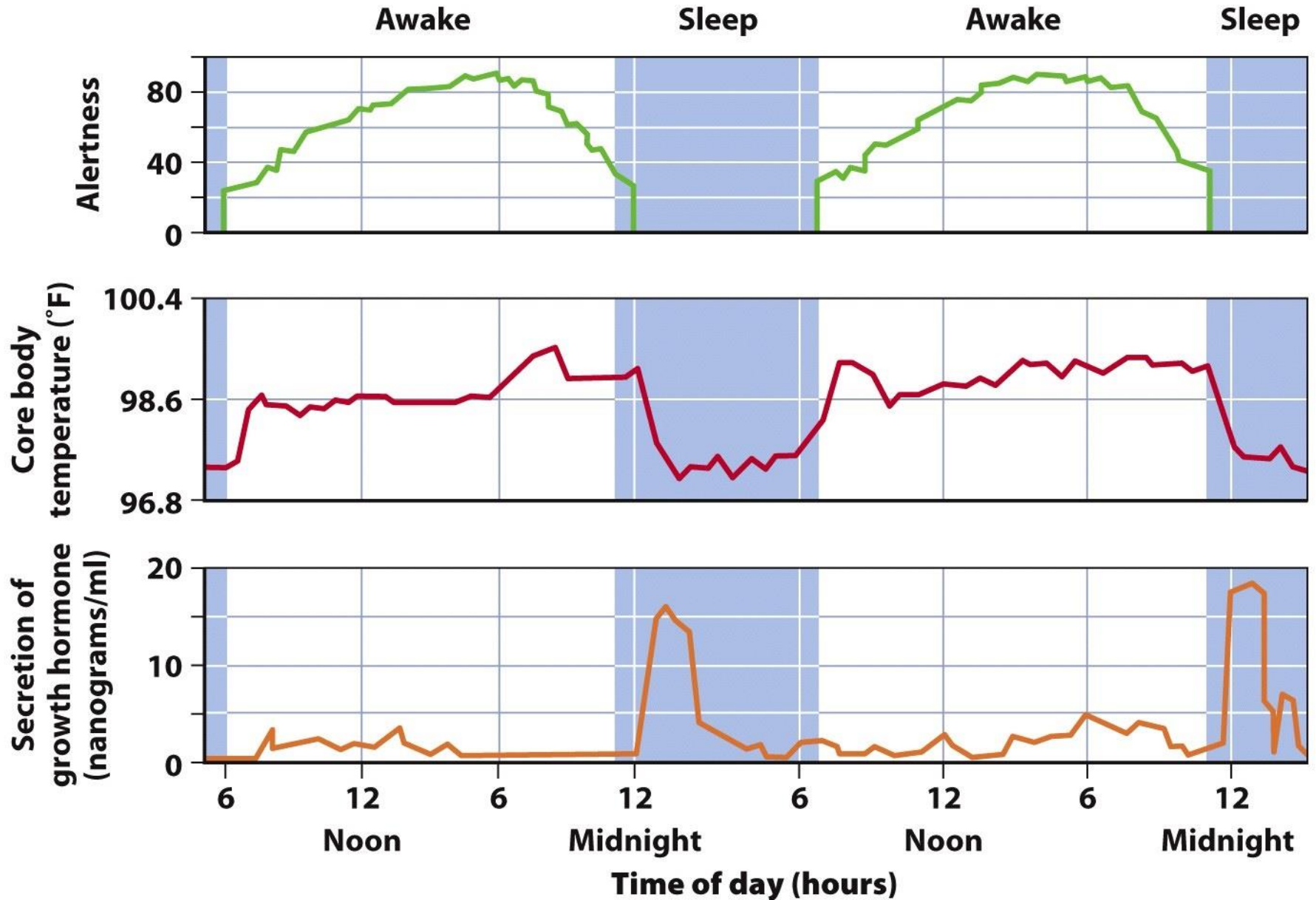
Questions?



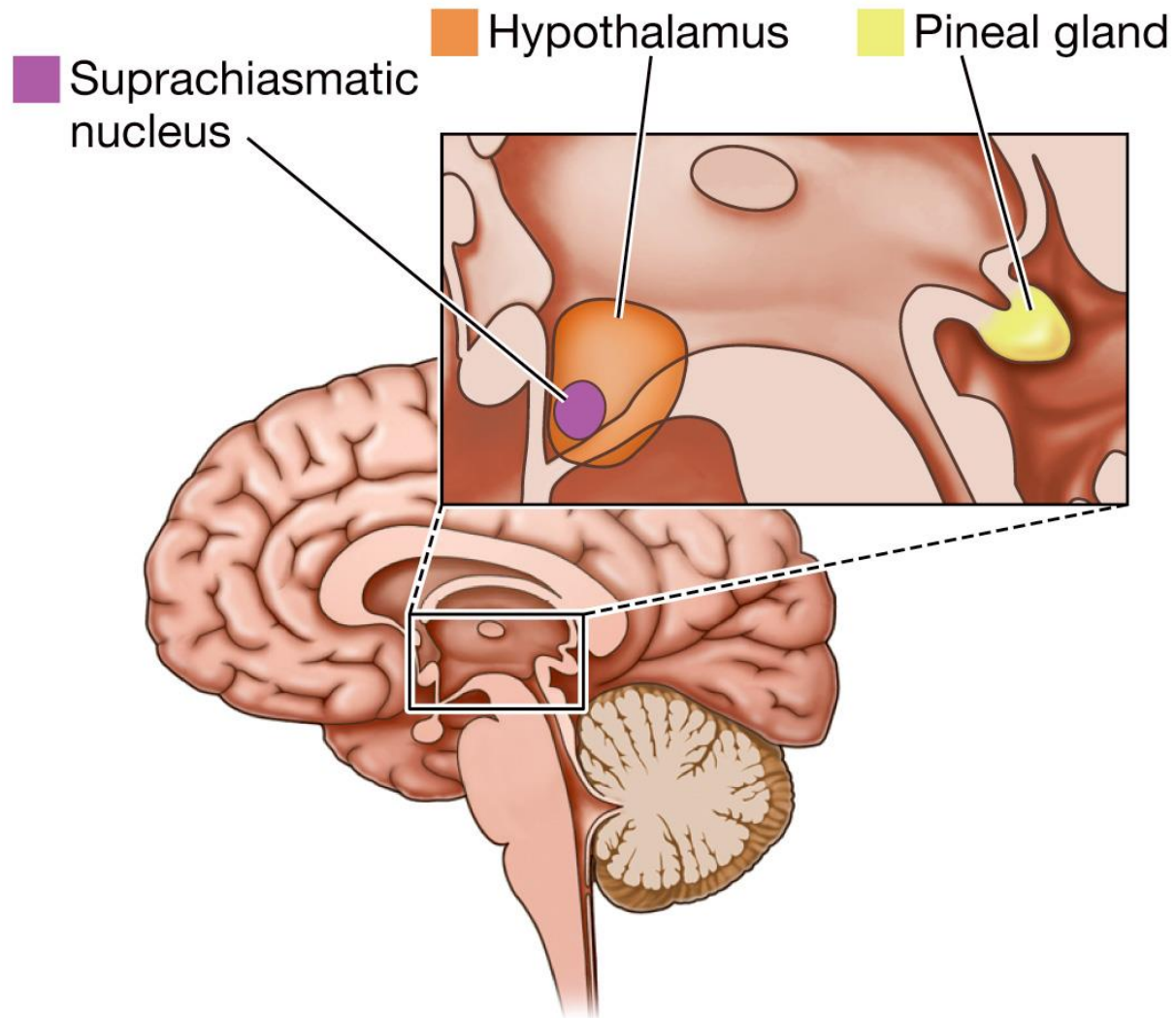
Sleep



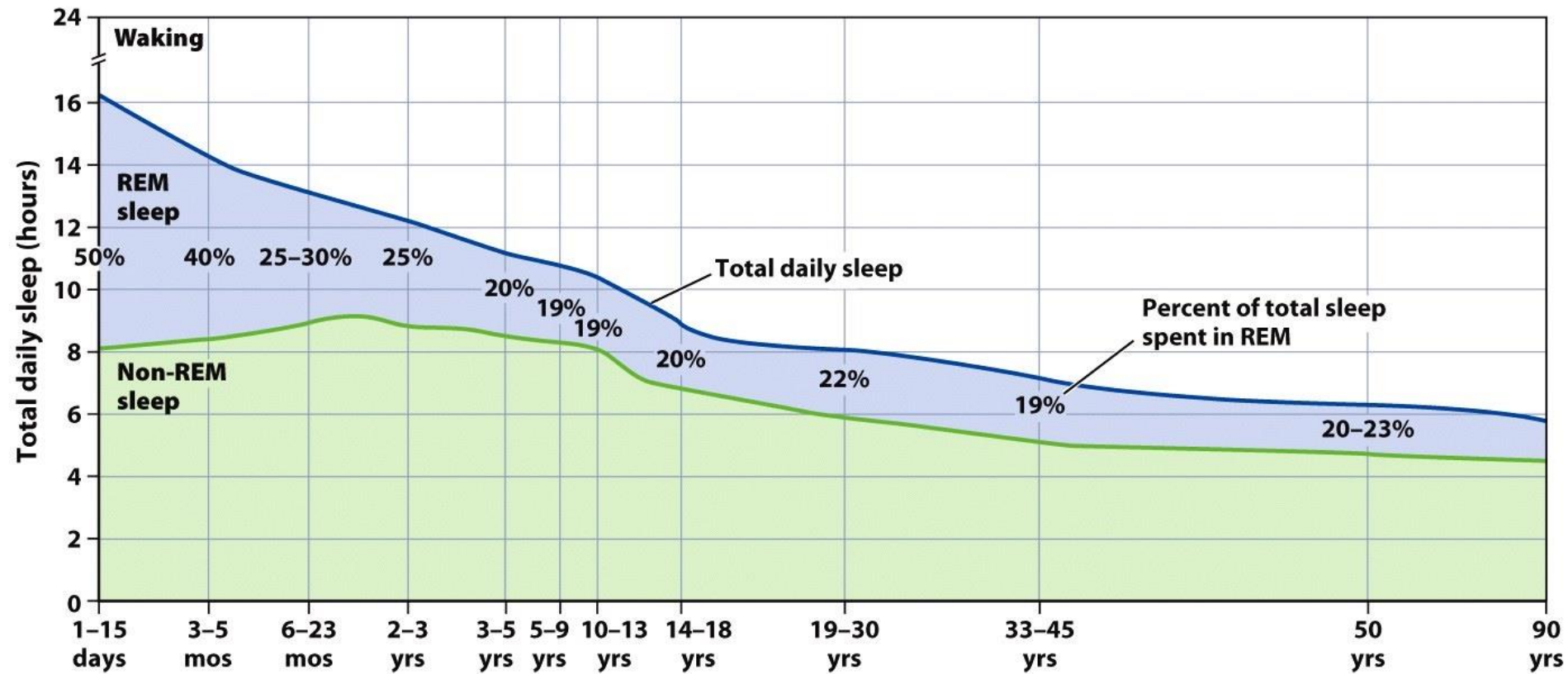
Sleep



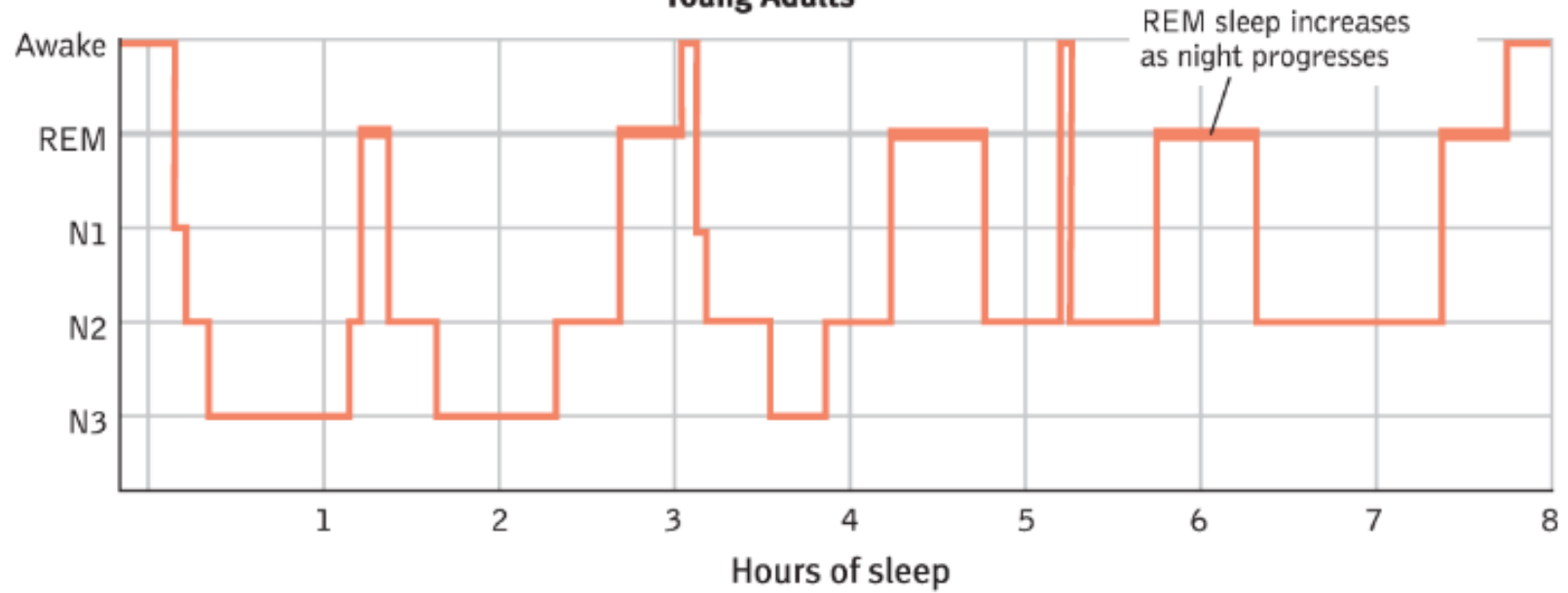
Sleep



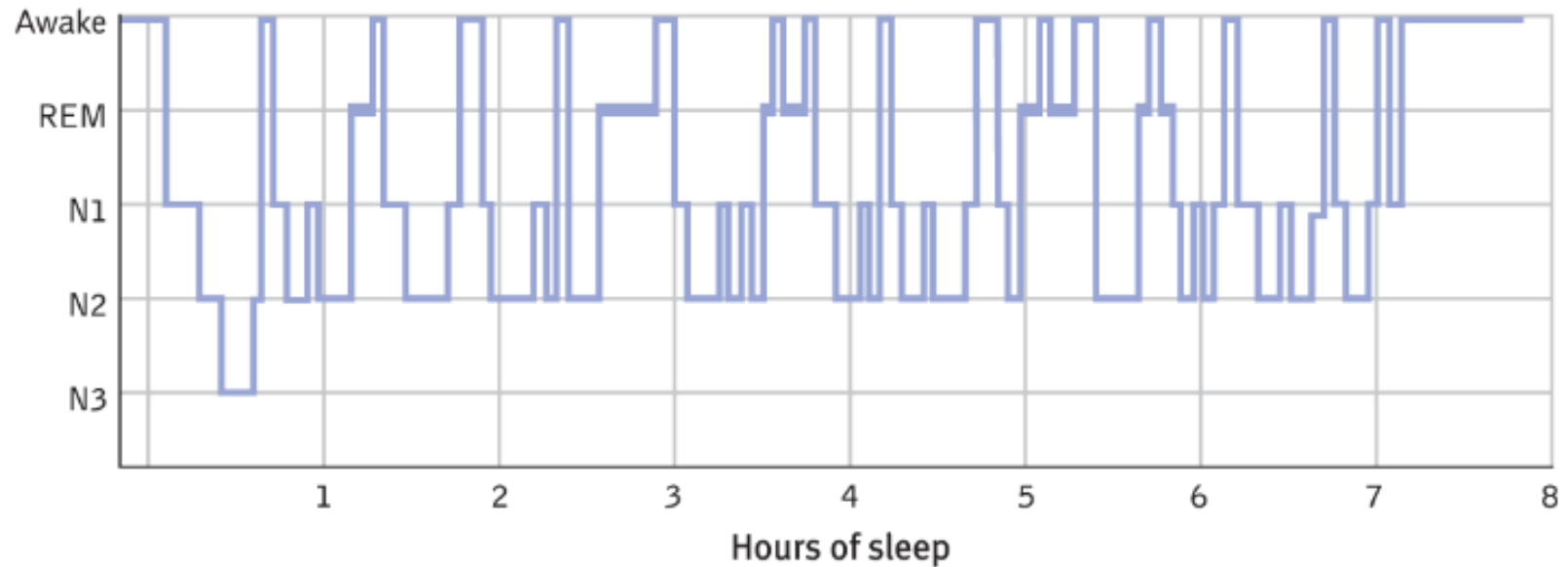
Sleep



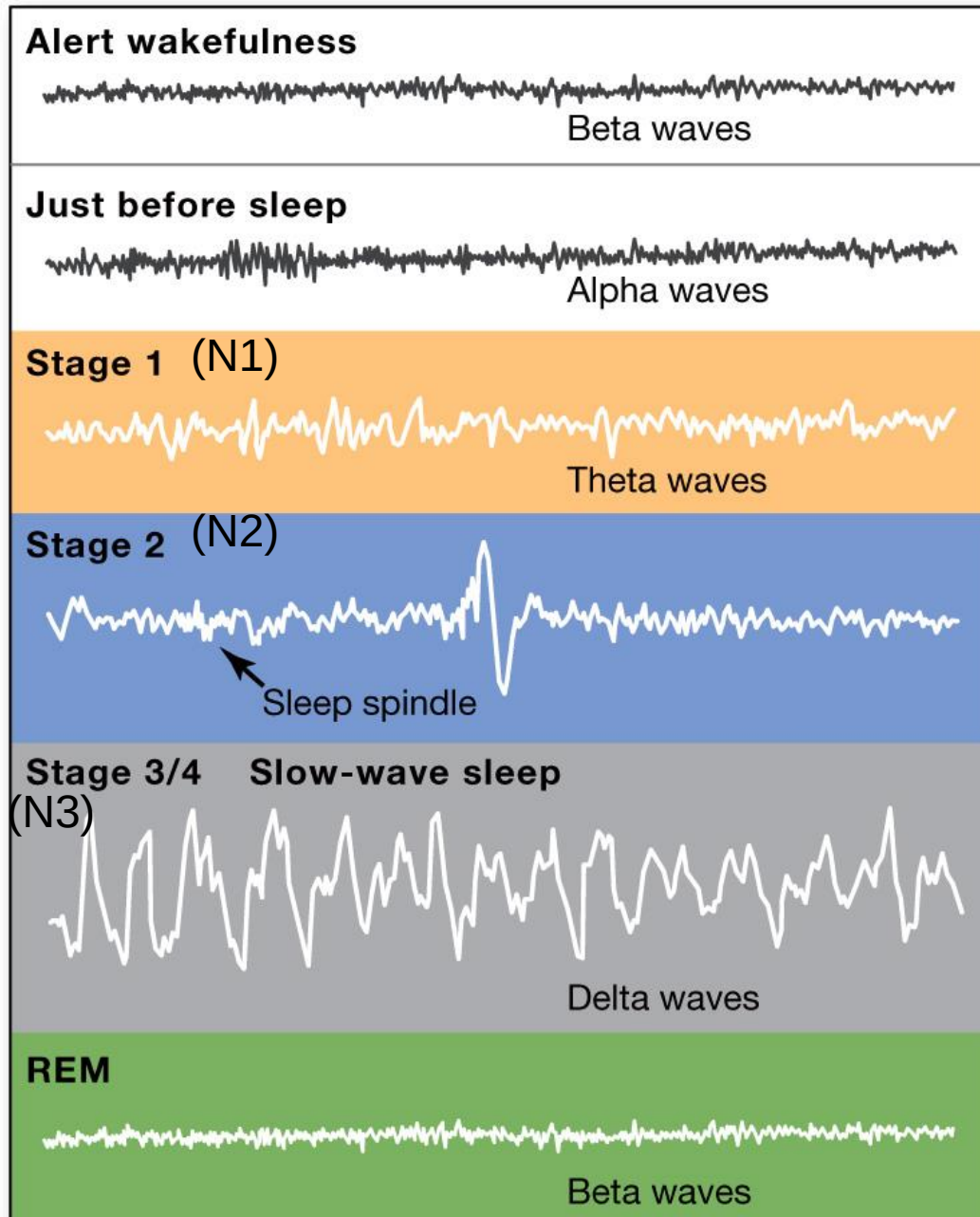
Young Adults



Older Adults



Sleep



**Awake with
mental activity**



**Beta
14-30 Hz**

**Awake and
resting**



**Alpha
8-13 Hz**

Sleeping



**Theta
4-7 Hz**

Deep sleep



**Delta
< 3.5 Hz**

1 second

Match the sleep stage (i–iii) with the cognitive experience (a–c):

i. N₁

ii. N₃

iii. REM

a. story-like dream

b. fleeting images

c. minimal awareness

Sleep

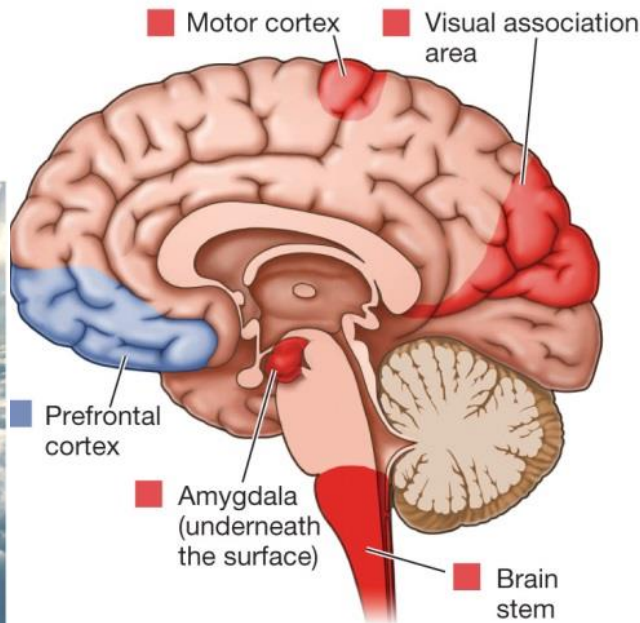
- Why do we sleep? -> theories about sleep
 - Restorative theory

Sleep

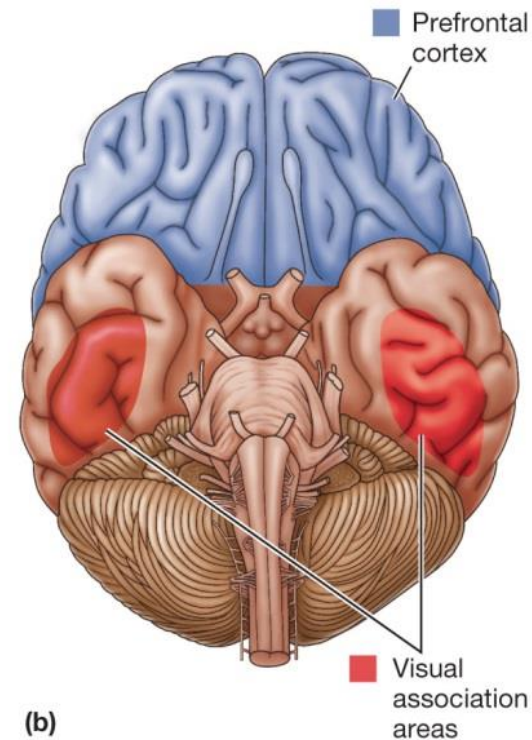
- Why do we sleep? -> theories about sleep
 - Restorative theory
 - Circadian rhythm theory
 - Facilitation of learning theory

Sleep

- **Dreams:** products of an altered state of consciousness in which images and fantasies are confused with reality
 - Non-REM dreams: dull
 - REM dreams: bizarre and intense



(a)

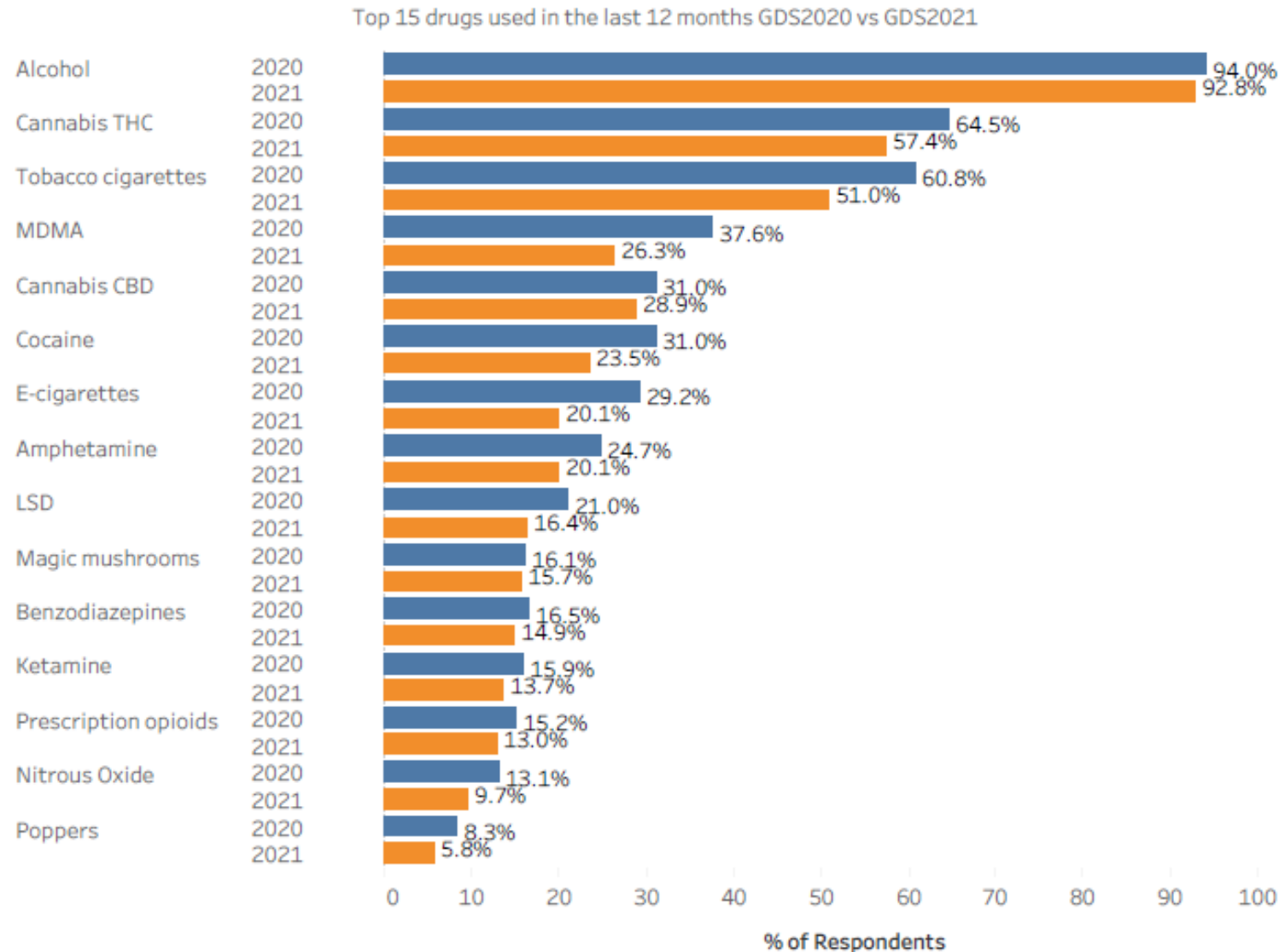


(b)

If you have any questions just say
“pspsps” and I’ll come to you



Altered Consciousness: drugs



Drugs

- **Tolerance:**
 - increasing amounts of a drug needed to achieve the intended effect
- **Addiction:**
 - drug use that remains compulsive despite its negative consequences
 - Physical and psychological dependence
- **Withdrawal:**
 - physiological and psychological state characterized by feelings of anxiety, tension, and cravings for the addictive substance

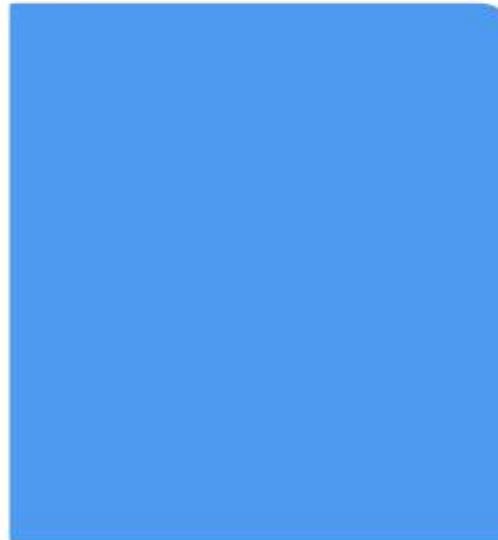
Many organs suffer damage despite the “tolerance”.



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Alcohol is a stimulant in small amounts but a depressant in large amounts.

62



True

38

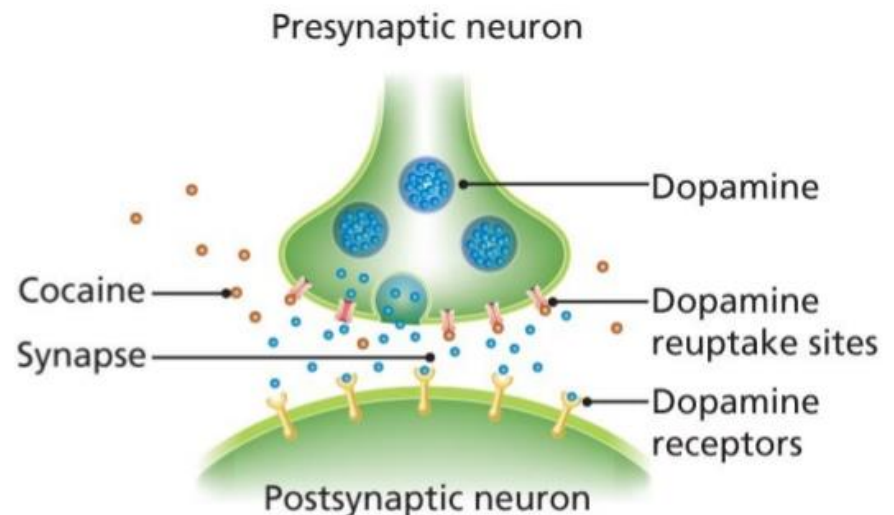


False

Drugs

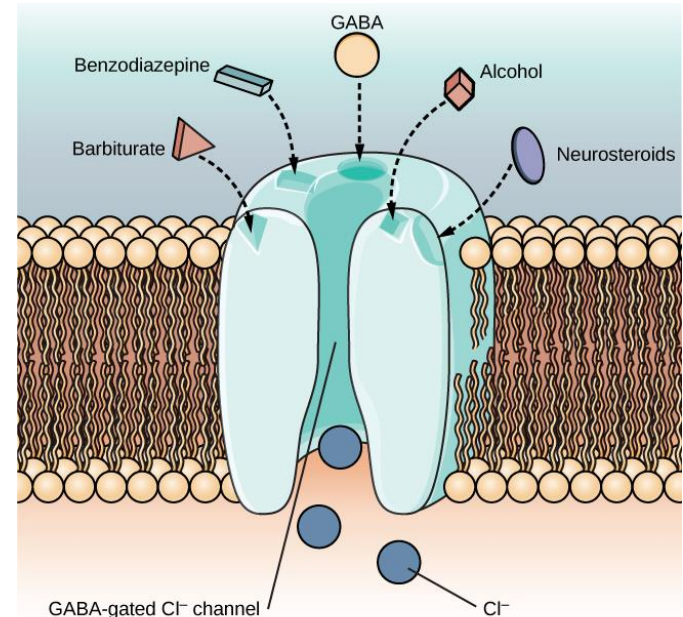
- **Stimulants**

- drugs that increase behavioral and mental activity and activate the sympathetic nervous system
 - Amphetamines
 - Methamphetamine
 - Cocaine
 - Nicotine
 - Caffeine



Drugs

- **Depressants:**
 - reduce behavioral and mental activity by depressing the central nervous system
 - alcohol
 - anti-anxiety drugs –benzodiazepines
 - barbiturates



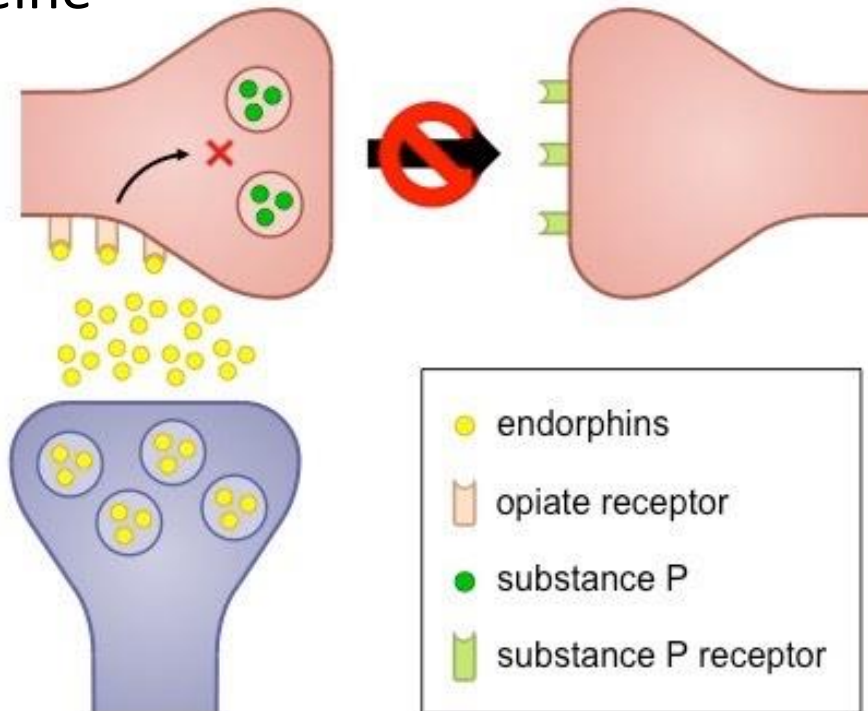
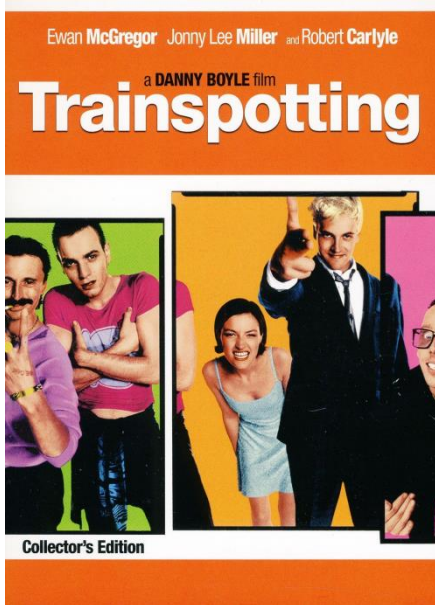
Drugs



<http://opioids.com/timeline>

- **Opiates (narcotics):**

- depress or slow down the central nervous system; relieve pain and suffering
 - Heroin
 - Morphine
 - Codeine



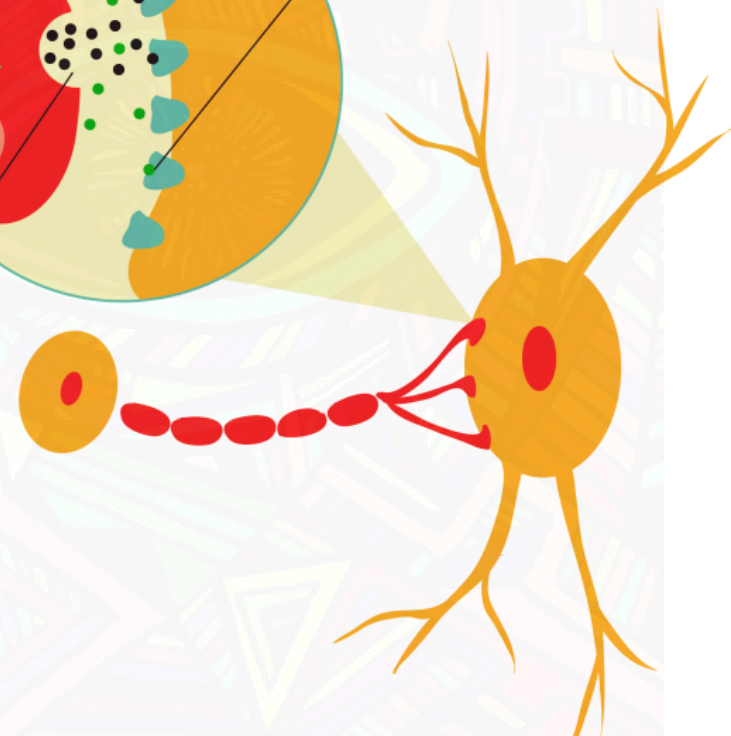
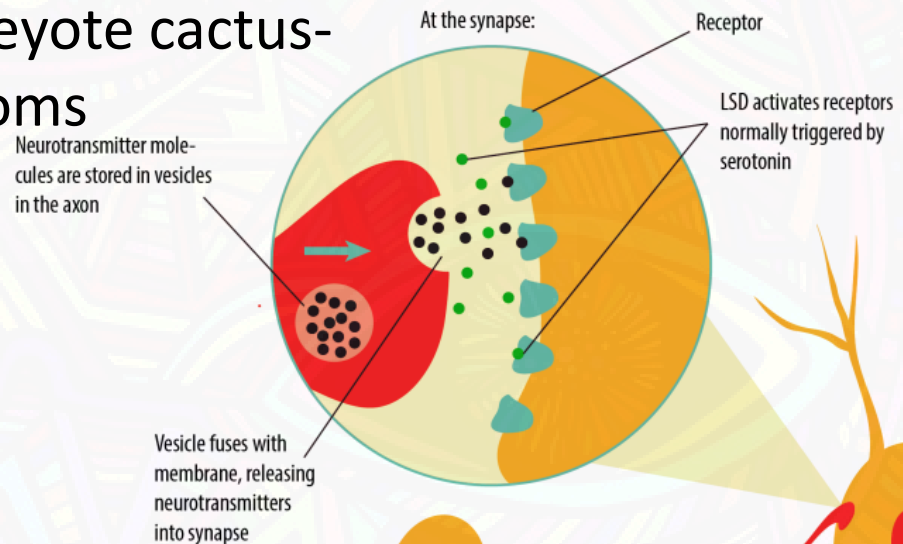
Substance P is a neurotransmitter involved in *pain responses*

Endorphins are released by the pituitary gland in order to *block pain perception*

Endorphins bind to opiate receptors located on the presynaptic membrane and *block the release of substance P*

Drugs

- **Hallucinogens (psychedelics):**
 - produce alterations in cognition, mood, and perception
 - LSD
 - Mescaline –from Peyote cactus-
 - psilocybin mushrooms



Drugs

LSD

Acid, Trips, Lucy

Serotonergic Psychedelic

+
SE

+
DA

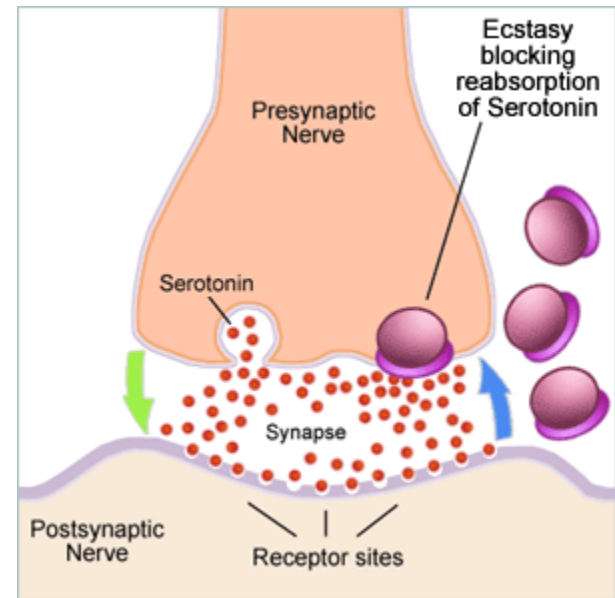
+
NE

+
GLU



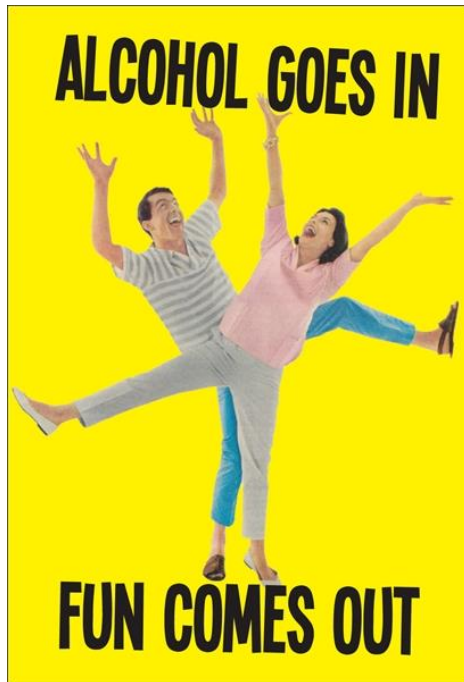
Drugs

- Many commonly used drugs do not fit into these four major categories (e.g., Marijuana, MDMA)



Drugs

- Drugs can have some positive effects...
- Yet, in the long run drugs are extremely harmful (to your body and your brain!)



when your lecturer asks if you have any questions



Can you repeat the part of the stuff
where you said all about the things?

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